

Business Management System (BMS) - Product Requirements Document & Technical Specifications.

Information Architecture & Navigation

The application employs a persistent left-hand sidebar navigation structure. The navigation items are segmented to differentiate department functions from standalone utility pages.

- **GOVERNANCE Section:** Houses the **Decisions** and **Approvals** pages.
- **OPERATIONS Section:** Houses the **Responsibilities** and **Actions** pages.
- **FINANCE Section:** Houses the **Expenses** page.
- **Standalone Navigation:** **Reports** and **Settings** exist as independent, standalone links located at the bottom of the navigation structure.

Functional Requirements

3.1 Dashboard Overview Metrics

The top section of the dashboard must display the following five key performance indicators (KPIs) in individual metric summary cards:

- **Open Decisions:** Total count of active decisions.
- **Pending Approvals:** Total count of approvals awaiting managerial action.
- **Active Responsibilities:** Total number of assigned operational roles or areas.
- **Open Actions:** Total count of individual tasks currently open.
- **Monthly Expenses:** A monetary display tracking the total expenses incurred during the current month.

3.2 Recent Decisions Log

A data table that lists recently logged decisions to maintain visibility. The table must include the following attributes:

- **Decision Name:** Text describing the specific action or decision.
- **Owner:** The specific individual responsible for managing or initiating the decision.
- **Date:** The date of the decision entry.
- **Status Label:** Clear badges indicating the current state (for example, Approved, Pending Approval, Draft).

3.3 Pending Approvals Component

A module displaying urgent requests that require direct intervention.

- **Item Types:** Must support various request types, such as Equipment Purchase Request, Budget Reallocation Proposal, and New Vendor Contract.
- **Metadata:** Each item must clearly display the author of the request and the submission date.
- **Interactions:** Each item must include two explicit action buttons: Approve and Reject.

3.4 Team Activity Feed

A chronological log documenting recent workspace events to foster team collaboration and transparency.

- **Event Details:** Must describe the actor, the action taken, and any relevant entity (for example, Sarah Johnson approved decision "Implement new customer onboarding process").
- **Time Indicator:** Must show a relative timestamp indicating when the action occurred (for example, 2 hours ago).

Page Functional Specifications

4.1 Decisions Page

The Decisions page serves as the database for tracking operational choices made within the organization.

- **Title:** Decisions
- **Subtitle:** Track and manage company decisions
- **Primary Action:** Add Decision button to initiate a new decision entry.
- **Search Bar:** Dynamic search input with the placeholder "Search decisions..."

4.2 Approvals Page

The Approvals page is restricted to managers and administrators who possess authorization to review, sign off on, or reject operational requests.

- **Title:** Approvals
- **Subtitle:** Review and approve pending decision requests
- **Pending Review Queue:** Displays request cards with Decision Title, Submitter Name, Est. Cost, and Justification Block along with interactive Approve, Reject, and Details buttons.

Historical Approvals Database:

5. Technical Architecture & Database Design

5.1 Entity-Relationship Diagram (ERD)

```
USERS ||--o{ DECISIONS : "submits / resolves"
USERS ||--o{ RESPONSIBILITIES : "owns"
USERS ||--o{ ACTIONS : "is assigned to"
USERS ||--o{ EXPENSES : "incurs"
USERS ||--o{ ACTIVITY_LOGS : "performs"
```

```
USERS {
  int id PK
  string first_name
  string last_name
  string email
  string role
  datetime created_at
}
```

```
DECISIONS {
  int id PK
  string title
  string description
  int submitted_by_user_id FK
  decimal estimated_cost
  string status
```

```
    datetime created_at
    datetime resolved_at
    int resolved_by_user_id FK
    string rejection_reason
}
```

```
RESPONSIBILITIES {
    int id PK
    string title
    string description
    int owner_user_id FK
    string status
    datetime created_at
}
```

```
ACTIONS {
    int id PK
    string title
    string description
    int assigned_to_user_id FK
    string status
    date due_date
    datetime created_at
}
```

```
EXPENSES {
    int id PK
    string title
    decimal amount
    date date
    string category
    int submitted_by_user_id FK
    string status
    datetime created_at
}
```

```
ACTIVITY_LOGS {
    int id PK
    int actor_user_id FK
```

```
    string action
    string target_type
    int target_id
    string details
    datetime created_at
}
```

5.2 Data Dictionary

Table 1: users

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the user.
first_name	VARCHAR(50)	NOT NULL	The first name of the user.
last_name	VARCHAR(50)	NOT NULL	The last name of the user.
email	VARCHAR(100)	UNIQUE, NOT NULL	Professional email address.
role	VARCHAR(20)	NOT NULL	System access level (Employee, Manager, Admin).
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp.

Table 2: decisions

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the decision.
title	VARCHAR(255)	NOT NULL	Short title of the decision.
description	TEXT	NOT NULL	Detailed scope of the action.
submitted_by_user_id	INT	FK -> users.id, NOT NULL	The user who submitted the decision request.
estimated_cost	DECIMAL(12, 2)	DEFAULT 0.00, NOT NULL	Estimated cost in USD.
status	VARCHAR(20)	NOT NULL	Lifecycle state: Draft, Pending Approval, Approved, Rejected.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date the decision was logged.
resolved_at	TIMESTAMP	NULLABLE	Date the decision was resolved.
resolved_by_user_id	INT	FK -> users.id, NULLABLE	The manager who resolved the decision.
rejection_reason	TEXT	NULLABLE	Populated only if status is Rejected.

Table 3: responsibilities

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the responsibility block.
title	VARCHAR(255)	NOT NULL	Title of the responsibility area.
description	TEXT	NOT NULL	Detailed operational expectations.
owner_user_id	INT	FK -> users.id, NOT NULL	The assigned owner of the duty.
status	VARCHAR(20)	NOT NULL	Current status (Active, Inactive).
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation date.

Table 4: actions

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the operational task.
title	VARCHAR(255)	NOT NULL	Task title or objective.
description	TEXT	NOT NULL	Detailed instruction

Column Name	Data Type	Constraints	Description
			steps.
assigned_to_user_id	INT	FK -> users.id, NOT NULL	The user assigned to complete the action.
status	VARCHAR(20)	NOT NULL	Resolution state (Open, In Progress, Completed).
due_date	DATE	NULLABLE	Operational deadline.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Task log creation time.

Table 5: expenses

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the expense transaction.
title	VARCHAR(255)	NOT NULL	Name/label for the expense.
amount	DECIMAL(12, 2)	NOT NULL	Expense cost in USD.
date	DATE	NOT NULL	Date the cost was incurred.

Column Name	Data Type	Constraints	Description
category	VARCHAR(50)	NOT NULL	Financial category (Hardware, Software, Travel, etc.).
submitted_by_user_id	INT	FK -> users.id, NOT NULL	The user who submitted the expense.
status	VARCHAR(20)	NOT NULL	Approval status (Pending, Approved, Rejected).
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Upload registration date.

Table 6: activity_logs

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the activity event.
actor_user_id	INT	FK -> users.id, NOT NULL	The user who performed the activity.
action	VARCHAR(50)	NOT NULL	Type of action (created, approved, rejected, deleted).
target_type	VARCHAR(50)	NOT NULL	The referenced table name (decisions,

Column Name	Data Type	Constraints	Description
			expenses, etc.).
target_id	INT	NOT NULL	The matching ID in the referenced table.
details	TEXT	NULLABLE	Description text compiled for the activity log feed.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of the activity.

6. Non-Functional Requirements

- **Data Integrity:** Actions approved or rejected on the Approvals page must immediately update corresponding statuses in the Decisions database and reflect on Dashboard KPI metrics.
- **Permission Model:** Access to the Approvals page must be restricted to users mapped to the Manager or Admin roles. Unauthorized access attempts must result in a 403 Forbidden screen.
- **Responsive Layout:** The sidebar navigation must scale dynamically, collapsing to a hamburger drawer menu on mobile displays to maximize working screen space for data grids.